Parallel stages: A combination of two or more stages with the same input and added outputs.

Parameter space: Target detection jargon. One parameter can be graphically interpreted as a *line*, two parameters a *plane*, three parameters a *space*, and more than three parameters a *hyperspace*.

Parseval's relation: Equation relating the energy in the time domain to the energy in the frequency domain.

Passband: The band of frequencies a filter is designed to pass unaltered.

Passive sonar: Detection of submarines and other undersea objects by the sounds they produce. Used for covert surveillance.

Phasor transform: Method of using complex numbers to find the frequency response of *RLC* circuits. Resistors, capacitors and inductors become R, $-j/\omega C$, and $j\omega L$, respectively.

Pillbox: Shape of a filter kernel used in image processing: circular region of a constant value surrounded by zeros.

Pitch: Human perception of the fundamental frequency of an continuous tone. See *timbre* for comparison.

Pixel: A contraction of "picture element." An individual sample in a digital image.

Point spread function (PSF): Imaging jargon for the impulse response.

Pointer: A variable whose value is the address of another variable.

Poisson statistics: Variations in a signal's value resulting from it being represented by a finite number of particles, such as: x-rays, light photons or electrons. Also called *Poisson noise* and *statistical noise*.

Polar form: Representing sinusoids by their magnitude and phase: $M\cos(\omega t + \phi)$, where M is the magnitude and ϕ is the phase. See *rectangular form* for comparison.

Pole: Term used in the Laplace transform and z-transform. When the s-domain or z-domain transfer function is written as one polynomial divided by another polynomial, the roots of the denominator are the *poles* of the system, while the roots of the numerator are the *zeros*.

Pole-zero diagram: Term used in the Laplace and z-transforms. A graphical display of the location of the poles and zeros in the s-plane or z-plane.

Precision: The error in a measurement or prediction that is not repeatable from trial to trial. Precision is determined by random errors. See *accuracy* for comparison.

Probability distribution function (pdf): Gives the probability that a *continuous* variable will take on a certain value.

Probability mass function (pmf): Gives the probability that a *discrete* variable will take on a certain value. See *pdf* for comparison.

Pulse response: The output of a system when the input is a pulse.

Quantization error: The error introduced when a signal is quantized. In most cases, this results in a maximum error of ±½ LSB, and an rms error of $1/\sqrt{12}$ LSB. Also called *quantization noise*.

Random error: Errors in a measurement or prediction that are not repeatable from trial to trial. Determines *precision*. See *systematic error* for comparison.

Radar: Radio Detection And Ranging. Echo location technique using radio waves to detect aircraft.

Real DFT: The discrete Fourier transform using only real (ordinary) numbers. A less powerful technique than the complex DFT, but simpler. See *complex DFT* for comparison.

Real FFT: A modified version of the FFT. About 30% faster than the standard FFT when the time domain is completely real (i.e., the imaginary part of the time domain is zero).

Real Fourier transform: Any of the members of the Fourier transform family using only real (as opposed to imaginary or complex) numbers. See *complex Fourier transform* for comparison.

Real part: The portion of a complex number that does not have the j term, such as 3 in $3 + 2j$. In the real Fourier transform, the *real* part refers to the part of the frequency domain that holds the amplitudes of the cosine waves, even though no j terms are present.

Real time processing: Processing data as it is acquired, rather than storing it for later use.